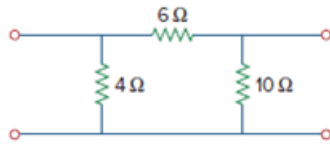


## Problem

Determine the  $h$  parameters for the circuit in Fig. 19.24.

**Figure 19.24**

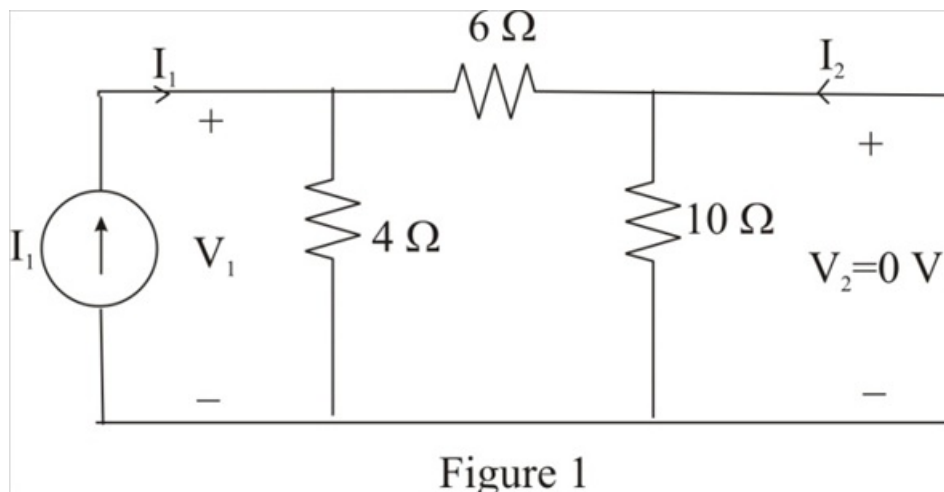


## Step-by-step solution

## Step 1 of 10

Refer to Figure 19.24 in the text book.

Connect the current source  $I_1$  as shown in Figure 1. Identify the currents and voltages in the circuit.



## Step 2 of 10

Consider the expression for the voltage  $V_1$ .

$$\begin{aligned} V_1 &= I_1 (4 \parallel 6) \\ &= I_1 \left( \frac{(4)(6)}{4+6} \right) \\ &= I_1 \left( \frac{24}{10} \right) \\ &= I_1 (2.4) \end{aligned}$$

Therefore, the expression for the voltage  $V_1$  is  $I_1 (2.4)$ .

## Step 3 of 10

Consider the expression for the current  $\mathbf{I_2}$ .

$$\begin{aligned}\mathbf{I_2} &= -\mathbf{I_1} \left( \frac{4}{4+6} \right) \\ &= -\mathbf{I_1} \left( \frac{4}{10} \right) \\ &= -\mathbf{I_1} (0.4)\end{aligned}$$

Therefore, the expression for the current  $\mathbf{I_2}$  is  $-\mathbf{I_1} (0.4)$ .

**Step 4** of 10

Determine the expression for the short circuit input impedance  $\mathbf{h_{11}}$ .

$$\mathbf{h_{11}} = \frac{\mathbf{V_1}}{\mathbf{I_1}}$$

Substitute  $\mathbf{I_1 (2.4)}$  for  $\mathbf{V_1}$ .

$$\begin{aligned}\mathbf{h_{11}} &= \frac{\mathbf{I_1 (2.4)}}{\mathbf{I_1}} \\ &= 2.4 \, \Omega\end{aligned}$$

Therefore, the expression for the short circuit input impedance  $\mathbf{h_{11}}$  is  $\boxed{2.4 \, \Omega}$ .

**Step 5** of 10

Determine the value of the short circuit forward current gain  $\mathbf{h_{21}}$ .

$$\mathbf{h_{21}} = \frac{\mathbf{I_2}}{\mathbf{I_1}}$$

Substitute  $-\mathbf{I_1 (0.4)}$  for  $\mathbf{I_2}$

$$\begin{aligned}\mathbf{h_{21}} &= \frac{-\mathbf{I_1 (0.4)}}{\mathbf{I_1}} \\ &= -0.4\end{aligned}$$

Therefore, the value of the short circuit forward current gain  $\mathbf{h_{21}}$  is  $\boxed{-0.4}$ .

**Step 6** of 10

Connect the current source  $\mathbf{I_2}$  as shown in Figure 2. Identify the currents and voltages in the circuit.

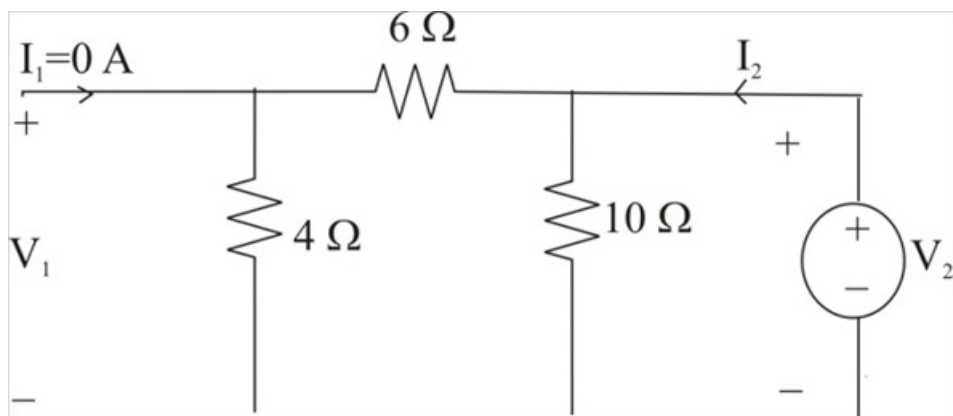


Figure 2

**Step 7** of 10

Consider the expression for the voltage  $V_2$ .

$$\begin{aligned} V_2 &= I_2 (10 \parallel (6+4)) \\ &= I_2 (10 \parallel 10) \\ &= I_2 \left( \frac{(10)(10)}{10+10} \right) \\ &= I_2 \left( \frac{100}{20} \right) \end{aligned}$$

$$V_2 = 10I_2$$

Therefore, the expression for the voltage  $V_2$  is  $I_2(5)$ .

**Step 8** of 10

Consider the expression for the voltage  $V_1$ .

$$\begin{aligned} V_1 &= I_2 \left( \frac{10}{10+6+4} \right) (4) \\ &= I_2 \left( \frac{10}{20} \right) (4) \\ &= I_2 (2) \end{aligned}$$

Therefore, the expression for the voltage  $V_1$  is  $I_2(2)$ .

**Step 9** of 10

Determine the value of the open circuit reverse voltage gain  $h_{12}$ .

$$h_{12} = \frac{V_1}{V_2}$$

Substitute  $I_2(2)$  for  $V_1$  and  $I_2(10)$  for  $V_2$ .

$$\begin{aligned} h_{12} &= \frac{I_2(2)}{I_2(10)} \\ &= 0.4 \end{aligned}$$

Therefore, the value of the open circuit reverse voltage gain  $h_{12}$  is 0.4.

**Step 10** of 10

Determine the expression for the open circuit output admittance  $h_{22}$ .

$$h_{22} = \frac{I_2}{V_2}$$

Substitute  $I_2(5)$  for  $V_2$ .

$$\begin{aligned} h_{22} &= \frac{I_2}{I_2(5)} \\ &= 200 \text{ mS} \end{aligned}$$

Therefore, the expression for the open circuit output admittance  $h_{22}$  is 200 mS.